Original Article

Superiority of single-photon emission computed tomography for predicting postoperative lung function after anatomical lung resection: a systematic review and meta-analysis

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Background: Accurate prediction of postoperative pulmonary function is crucial for surgical decision-making in patients undergoing anatomic lung resection for non-small cell lung cancer (NSCLC). Traditional assessment methods based on planar ventilation/perfusion (V/Q) scintigraphy or anatomical segment calculation (ASC) are limited by low spatial resolution and simplified assumptions of uniform segmental volume and contribution. The use of hybrid single-photon emission computed tomography/computed tomography (SPECT/CT) has emerged as a modern alternative for three-dimensional functional assessment, but its comparative accuracy remains uncertain. Therefore, this study aimed to compare the accuracy of SPECT/CT with planar V/Q scintigraphy and ASC in predicting postoperative pulmonary function after anatomic lung resection.

Methods: We systematically searched PubMed, Embase, and the Cochrane Library through June 2025 for studies comparing SPECT/CT with conventional methods (planar V/Q scan or ASC) for predicting postoperative pulmonary function after anatomic lung resection. Eligible studies included adults with lung cancer reporting both predicted and observed postoperative forced expiratory volume in one second (FEV_1) and/or diffusing capacity for carbon monoxide (DLCO) values. Statistical analysis was performed using Bland-Altman plots, linear regression, and bar-plot comparison of absolute differences between predicted and observed values. The protocol was registered in PROSPERO (CRD420251103221).

Results: Five studies comprising 424 patients met inclusion criteria. Across cohorts, mean age ranged from 66 to 68 years, and lobectomy represented the predominant type of surgical resection. SPECT/CT demonstrated the smallest bias and narrowest limits of agreement for both FEV_1 [mean difference (MD) =−4.5] and DLCO (MD =−2.2), outperforming ASC (MD =−4.2) and planar V/Q (MD =−12.9). Correlation between predicted and observed values was strongest with SPECT/CT (R^2=0.89 for FEV_1; R^2=0.91 for DLCO), intermediate with ASC, and weakest with V/Q. Absolute prediction errors were lowest with SPECT/CT, indicating superior agreement and precision. Overall evidence quality by Grading of

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Recommendation, Assessment, Development, and Evaluations (GRADE) was rated as moderate for SPECT/CT and low for conventional methods.

Conclusions: In patients with NSCLC undergoing anatomic lung resection, preoperative SPECT/CT provides the most accurate prediction of postoperative FEV_1 and DLCO values than planar V/Q scintigraphy or ASC. Our findings suggest that SPECT/CT should be considered the preferred modality for functional evaluation before lung resection, particularly in patients with borderline pulmonary reserve. Future multicenter prospective studies should validate standardized thresholds for surgical eligibility.

Keywords: Lung cancer; anatomic lung resection; pulmonary function tests (PFTs); predicted postoperative lung function; single-photon emission computed tomography/computed tomography (SPECT/CT)

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Introduction

Lung cancer remains the leading cause of cancer-related mortality worldwide (1). In patients with early-stage disease, surgical resection offers the best chance for cure (2). In clinical practice, however, these patients represent significant decision-making challenge with higher level of frailty, advanced age, and significant burden of comorbidities, including chronic obstructive pulmonary disease, with impaired lung function (3). In light of this, parenchyma sparing procedures such as anatomical segmentectomy or wedge resection have been brought to the forefront of the clinical practice by several recent trials, especially for small peripheral tumors (4-6). As such, through preoperative evaluation of patient with accurate prediction of pulmonary function after lung resection for cancer is crucial for making the best decision regarding local treatment, whether it is surgery or radiotherapy (7,8).

Accurate preoperative prediction of the impact of the anatomical lung resection on postoperative pulmonary function in patients with lung cancer is a critical component of preoperative evaluation (9-11). Forced expiratory volume in one second (FEV_1) and diffusing capacity for carbon monoxide (DLCO) are widely used parameters for determining surgical risk and guiding the extent of resection (12,13). Traditionally, postoperative prediction has been based on planar scintigraphy, also known as quantitative ventilation/perfusion scan (V/Q scan) or on the anatomical segment calculation (ASC) method (14,15). Although both approaches are well established, they are limited by simplified assumptions of uniform segmental volume and contribution and low spatial resolution of planar scintigraphy imaging, which introduces error in assessment of functional reserve, especially in patients with heterogeneous lung disease (16,17).

In recent years, single-photon emission computed tomography/computed tomography (SPECT/CT) for spatial resolution has been introduced as an advanced alternative, allowing three-dimensional evaluation of regional ventilation and perfusion with improved anatomical correlation (18). Despite its potential advantages, the clinical value of SPECT/CT compared with conventional methods remains unclear (19). Some reports suggest that SPECT/CT provides superior accuracy in predicting postoperative lung function, while others have found overlapping results with planar scintigraphy or ASC (20,21). The lack of large-scale comparative studies and methodological heterogeneity across available publications contributes to persistent uncertainty. Moreover, it is unclear whether SPECT/CT offers a consistent benefit across different patient populations, particularly among those with preserved lung function or borderline operability.

In view of these controversies, we conducted a systematic review and meta-analysis to evaluate whether preoperative SPECT/CT provides more accurate predictions of postoperative lung function than conventional approaches in lung cancer patients undergoing anatomical lung resection. We present this article in accordance with the PRISMA reporting checklist (available at https://jtd.amegroups.com/article/view/10.21037/jtd-2026-1-0155/rc).

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Methods

Protocol and registration

The study protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO) under registration number CRD420251103221.

Search strategy and study selection

We systematically searched PubMed, Embase and Cochrane Library databases with the cutoff date of June 30, 2025. The search strategy included the terms ("Lung Neoplasms" OR "lung cancer" OR "lung adenocarcinoma" OR "non-small cell lung cancer" OR NSCLC OR "squamous cell lung carcinoma" OR "lung carcinoma") AND ("SPECT/CT" OR "single-photon emission computed tomography" OR "hybrid imaging" OR "nuclear imaging" OR "functional imaging" OR "quantitative SPECT" OR "regional lung function assessment"). Two authors (L.M.D. and A.S.P.I.) independently conducted the search, imported results into Rayyan Software (Qatar Computing Research Institute, Qatar Foundation), and screened the studies. After the exclusion of duplicates and titles/abstracts unrelated to the clinical question, the eligibility of each remaining study was assessed based on the review of the articles' full text publication. Disagreements were solved by a third author (R.E.N.N.O).

Eligibility criteria

Inclusion criteria for studies were: (I) randomized controlled trials (RCTs) or observational studies; (II) enrolling adult patients with lung cancer and borderline lung function tests undergoing lung anatomical resection; (III) studies comparing SPECT/CT-based prediction of postoperative FEV_1 and/or DLCO to conventional methods: quantitative ventilation/perfusion (QVQ) scan and/or ASC; (IV) reporting at least one outcome of interest.

Studies were excluded based on the following criteria: (I) case reports, reviews, editorials, or conference abstracts. (II) Studies not reporting both predicted postoperative and observed postoperative lung function. (III) Pediatric population (IV) non-anatomical lung resections (e.g., wedge resection only).

Data extraction

Two authors (L.M.D. and A.S.P.I.) independently extracted data from the included studies using a standardized format. The collected variables included: study identifying information (title, year of publication, study design, study period and country), sample size, patient characteristics (age and gender), preoperative pulmonary function parameters (FEV_1 and DLCO), laterality of the surgical resection, and specific lobectomy definition (upper lobe, middle lobe or lower lobe).

Outcomes

The primary outcomes were predicted and observed postoperative FEV_1 and DLCO.

Statistical analysis

We used the Bland Altman method to evaluate the accuracy

Highlight box

Key findings

- In this systematic review and meta-analysis of five studies including 424 patients, single-photon emission computed tomography/computed tomography (SPECT/CT) showed the best performance for predicting postoperative pulmonary function after anatomic lung resection.
- SPECT/CT demonstrated the smallest bias, the narrowest limits of agreement, the strongest correlations between predicted and observed values, and the lowest absolute prediction errors for both forced expiratory volume in one second (FEV_1) and diffusing capacity for carbon monoxide (DLCO).

What is known and what is new?

- Conventional methods such as planar ventilation/perfusion (V/Q) scintigraphy and anatomical segment calculation (ASC) are widely used for preoperative functional assessment, but both rely on simplified assumptions and may be inaccurate in patients with heterogeneous lung disease.
- This study provides a head-to-head comparison showing that SPECT/CT is more accurate than both planar V/Q scintigraphy and ASC for predicting postoperative FEV_1 and DLCO after anatomic lung resection. The advantage of SPECT/CT was particularly evident for DLCO, for which conventional methods showed greater imprecision.

What is the implication, and what should change now?

- SPECT/CT should be considered the preferred modality for preoperative functional assessment in patients undergoing anatomic lung resection for non-small cell lung cancer, particularly in borderline surgical candidates.
- Where SPECT/CT is unavailable, ASC remains an acceptable alternative, but its limitations should be recognized. Future prospective multicenter studies should standardize imaging protocols and validate decision thresholds for surgical eligibility.

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of different prediction modalities for the postoperative pulmonary function test (PFT) values and agreement between predicted and observed values. A smaller mean difference (MD), approaching zero, indicates better agreement, while larger differences reflect poorer concordance. The graph includes the MD (predicted minus observed) and the corresponding standard deviation (SD) of these differences. Limits of agreement were determined using the 95% confidence interval (CI), providing a visual representation of the degree of agreement. We also used linear regression which evaluates both the direction (positive or negative) and the strength of the correlation between variables. The strength of association is quantified using the coefficient of determination (R^2). Values of R^2 of 1 indicate a 100% correlation, whereas values approaching 0 indicate no correlation. In general, an R^2 above 0.7 is accepted as a strong correlation. Finally, a bar plot was generated to display the absolute differences between predicted and observed values. By using the absolute value of the differences, the analysis disregards whether the predicted value was higher or lower, focusing instead on the magnitude of the discrepancy. The plot compares the three main modalities across the variables of FEV_1 and DLCO assessment. In this context, taller bars indicate greater discrepancies and, therefore, poorer diagnostic performance, whereas shorter bars represent better agreement. All statistical analyses were performed in The R Project for Statistical Computing software (version 4.4.3).

Data synthesis

All statistical analyses were conducted using The R Project for Statistical Computing software (version 4.4.3), with the "dplyr" package for data manipulation and "ggplot2" for graph customization. A Bland-Altman plot was generated to evaluate agreement between observed and predicted values of FEV_1 and DLCO across various diagnostic modalities. Linear regression was applied to assess positive or negative correlations between these observed and predicted values, quantified by the R^2 coefficient. Additionally, bar plots incorporating 95% CIs were created to facilitate comparisons among the diagnostic approaches.

Quality assessment and risk of bias

The overall quality of the evidence was assessed following the Grading of Recommendation, Assessment, Development, and Evaluations (GRADE) guidelines (22). Non-randomized retrospective studies were categorized as having very low, low, moderate, or high-quality evidence, depending on factors such as risk of bias, result inconsistency, imprecision, publication bias, and the magnitude of treatment effects.

The risk of bias for diagnostic accuracy studies was assessed using the Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2) tool. This instrument evaluates risk of bias and applicability across four key domains: (I) patient selection, (II) index test, (III) reference standard, and (IV) flow and timing. Each domain was rated as having a low, high, or unclear risk of bias (23-25). Applicability concerns were also judged within the first three domains. Two independent authors performed the assessments (L.M.D. and A.S.P.I.), and any disagreements were resolved through consensus with the senior author (R.E.N.N.O.).

Results

Study characteristics

Figure 1 shows the PRISMA flow diagram of the study selection process. The initial search yielded 2,096 results. After deduplication and evaluation of title and abstract, 2,072 articles were excluded, remaining 24 articles were assessed in full texts. After this, 19 articles were excluded as not meeting inclusion criteria. Five studies met all the eligibility criteria and were included in the final analysis (12,26-29).

Baseline characteristics

Of the five included studies, four directly compared SPECT/CT, QVQ scan, and ASC, while one study assessed SPECT/CT alone. Consequently, pairwise analyses of SPECT/CT versus V/Q scan and SPECT/CT versus ASC were performed using four studies each, whereas absolute accuracy metrics for SPECT/CT incorporated all five publications.

Across the included cohorts (n=424), the mean patient age ranged from 66 to 68 years, and the proportion of male participants varied between 48% and 69%. The weighted average preoperative FEV_1 was 72.7%, and the median preoperative DLCO was 70.7%. All studies were single-center observational series conducted in Israel, Japan, Germany, South Korea, and Italy between 2004 and 2024.

Regarding surgical characteristics, lobectomy was the most frequent surgical resection type, representing

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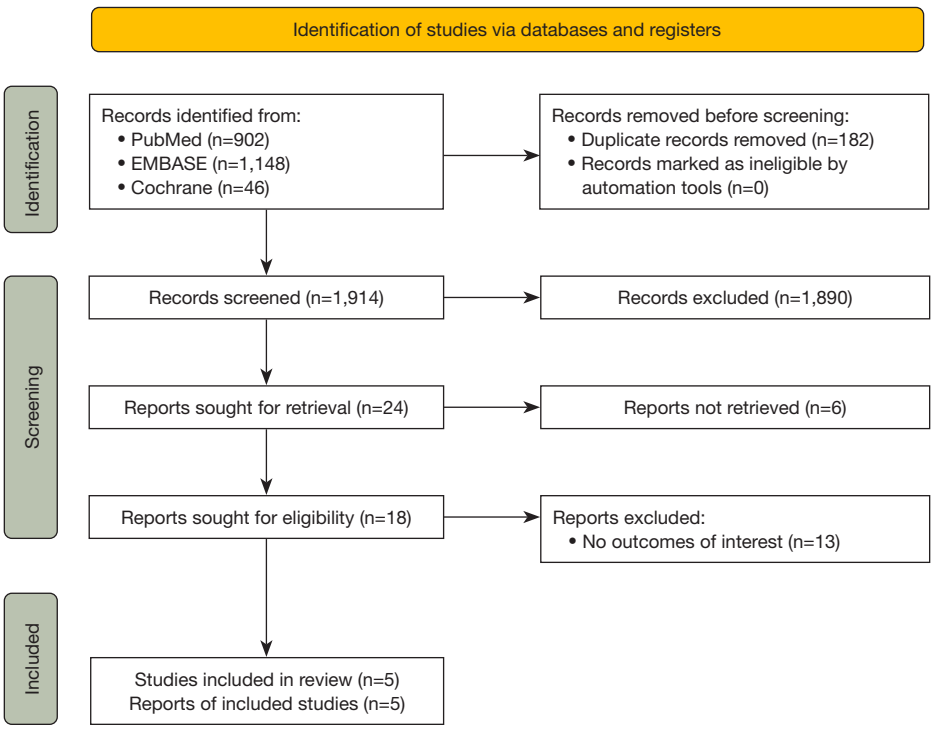


Figure 1 Study selection (PRISMA flow diagram).

41.6% to 100% of cases depending on the cohort, while bilobectomy and wedge resection were less common (≤10%). Only a small proportion of patients underwent pneumonectomy (≤6%). Laterality distribution showed a slight predominance of right-sided resections (55%) compared with left-sided resections (45%).

The detailed baseline characteristics of the included studies are summarized in *Table 1* and Table S1.

Bland-Altman analysis

SPECT/CT demonstrated the best agreement between predicted and observed postoperative FEV1 and DLCO values. For DLCO, SPECT/CT showed the MD closest to zero (MD =−2.2) and the lowest SD (2.77), indicating minimal bias and the highest precision among all methods (*Figure 2A*). This was visually reflected by the narrowest dispersion of points and the tightest limits of agreement, suggesting a high degree of concordance. For FEV_1, SPECT/CT also showed a relatively small MD (MD =−4.54; *Figure 2B*). In contrast, the QVQ scan showed the largest MD and the widest dispersion for both DLCO (MD =−12.95; SD: 24.87; *Figure 2C*) and FEV_1 (MD =−6.65; SD: 4.85; *Figure 2D*), indicating greater error and lower precision. The ASC method showed a smaller MD for DLCO (MD =−4.2; SD: 6.68; *Figure 2E*) than planar scintigraphy, suggesting better overall accuracy. However, both ASC plots (DLCO and FEV_1; *Figure 2E,2F*) suggested proportional bias, with increasing disagreement between predicted and observed values at higher predicted values. This pattern reduces the reliability of the ASC method, particularly in patients with preserved pulmonary function.

Linear regression modeling

A linear regression model assessed the relationship between predicted PPO and observed values. SPECT/CT showed the strongest correlation for both DLCO and FEV_1 (*Figure 3A,3B*, respectively), indicating strong predictive performance. QVQ demonstrated a moderate correlation for DLCO and FEV_1 (*Figure 3C,3D*, respectively). The ASC method showed weaker performance for DLCO but comparable performance for FEV_1 (*Figure 3E,3F*).

Bar plot

A bar plot was generated to display the absolute differences between predicted and observed values (*Figure 4*),

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Table 1 Baseline characteristics of the included studies

First author, year	Time frame	Study design	Country of origin	Sample size, n	Male	Mean age (years)	Preoperative FEV_1 (%)	Preoperative DLCO (%)	Laterality		Type of surgeical resection, n (%)			
									R	L	Pneumonectomy	Lobectomy	Bilobectomy	Wedge
Arnon-Sheleg, 2020 (12)	2016–2018	P-Obs	Israel, single-center	60	51 (68%)†	68.1†	74.5±18	68.9±17	31 (52%)	29 (4%)	0 (0.0)	60 (100.0)	0 (0.0)	0 (0.0)
Kawai, 2020 (26)	2018	R-Obs	Japan, single-center	30	20 (66%)	66	NA	NA	21 (70%)	9 (3%)	0 (0.0)	28 (93.3)	2 (6.7)	0 (0.0)
Moneke, 2024 (27)	NA	P-Obs	Germany, single-center	21	10 (48%)	NA	NA	NA	9 (42%)	12 (5%)	6 (28.6)	14 (66.7)	1 (4.8)	0 (0.0)
Na, 2020 (28)	2015–2018	R-Obs	South Korea, single-center	167	115 (69%)	66.6	84.28±18.87	72.45±14.06	NA	NA	15 (5.5)	114 (41.6)	12 (4.4)	26 (9.5)
Mineo, 2006 (29)	2004–2005	P-Obs	Italy, single-center	39	NA	67	59.41±5.44	NA	NA	NA	11 (28.2)	28 (71.8)	0 (0.0)	0 (0.0)

†, data extracted from the whole study sample. DLCO, diffusing capacity for carbon monoxide; FEV_1, forced expiratory volume in one second; L, left; NA, not applicable; P-Obs, prospective observational study; R, right; R-Obs, retrospective observational study.

demonstrating that QVQ had the largest discrepancies, particularly for DLCO, indicating a poor prediction accuracy for postoperative values. ASC showed intermediate accuracy, with higher differences for FEV_1 than DLCO. In contrast, SPECT/CT consistently showed the lowest discrepancies for both FEV_1 and DLCO values, supporting its value as the most accurate method for predicting postoperative function.

Quality assessment

According to the GRADE assessment, agreement between predicted and observed postoperative function using SPECT/CT was rated as moderate-certainty evidence for both DLCO and FEV_1. Agreement for ASC and QVQ was rated as low-certainty evidence for DLCO and for FEV_1.

Imprecision affected the QVQ results, particularly for DLCO, given the wide dispersion and broad limits of agreement, indicating greater error. Inconsistency was identified for the ASC method due to proportional bias, whereby discrepancies increased at higher predicted values (for both DLCO and FEV_1). Risk of bias was judged as some concerns across all methods because the evidence was derived from observational studies. Indirectness was minimal, as the reference standard (observed pulmonary function) directly reflects the clinical target outcome. A complete GRADE evidence profile is provided in Table S2.

Risk of bias and publications bias

As shown in *Figure 5*, the risk of bias was assessed for all included studies using the QUADAS-2. Three studies were assessed as having a low risk of bias, while two others demonstrated a moderate risk of bias.

Discussion

In this systematic review and meta-analysis of five studies including 424 patients, we compared methods for predicting postoperative pulmonary function in patients undergoing anatomic lung resection. Our main findings were as follows: (I) SPECT/CT demonstrated superior accuracy compared with both QVQ scan and ASC for estimating postoperative FEV_1 and DLCO; (II) Bland-Altman analysis revealed the lowest bias and dispersion with SPECT/CT, while QVQ scan showed the largest discrepancies; and (III) ASC showed intermediate performance with proportional bias at higher predicted values. Linear regression and bar-plot analyses

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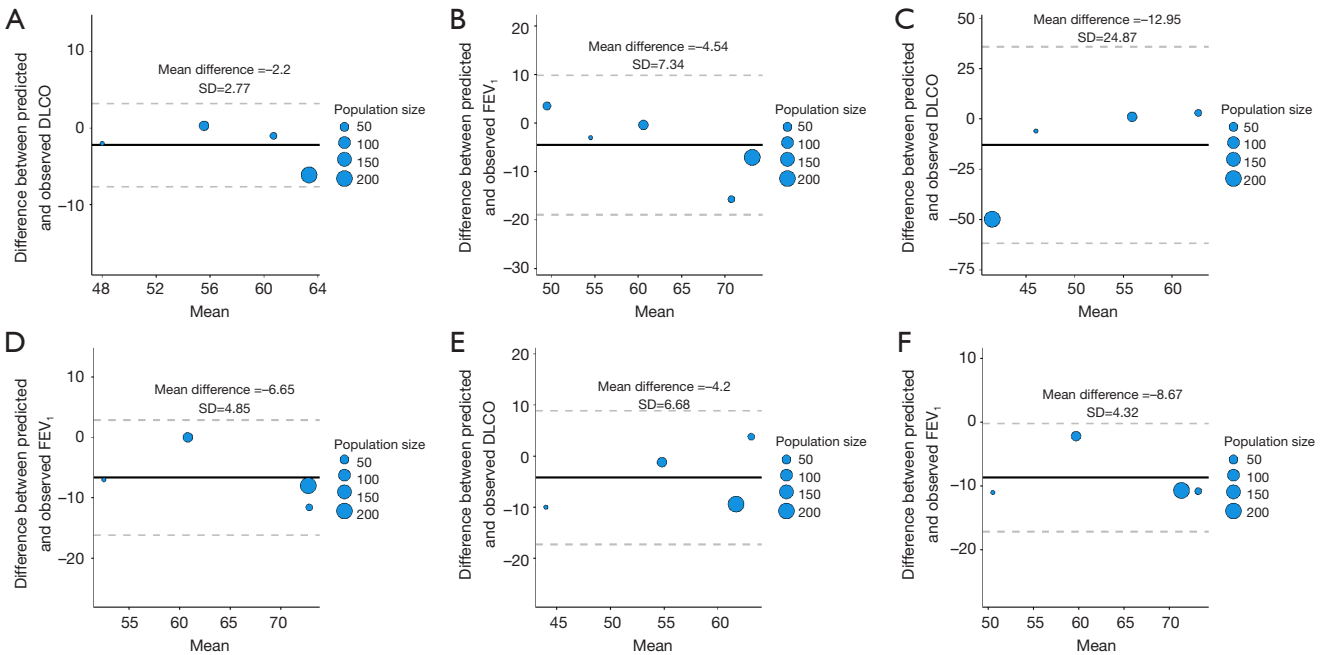


Figure 2 Bland-Altman plots demonstrated that SPECT/CT had the lowest bias and the narrowest limits of agreement, indicating the highest concordance with observed DLCO and FEV_1 (C,F). SC showed intermediate agreement (B,E) while V/Q scan exhibited the widest dispersion and bias (A,D). DLCO, diffusing capacity for carbon monoxide; FEV_1, forced expiratory volume in one second; SC, segment counting; SD, standard deviation; SPECT/CT, single-photon emission computed tomography/computed tomography; V/Q, ventilation/perfusion.

confirmed these results, showing the strongest correlations and smallest absolute prediction errors with SPECT/CT across both endpoints.

These results align with clinical reasoning. By integrating three-dimensional functional and anatomical information, SPECT/CT captures regional heterogeneity in ventilation and perfusion that planar imaging on the QVQ cannot resolve and that ASC assumes to be uniform. In patients with non-small cell lung cancer (NSCLC), heterogeneous parenchymal involvement is frequent, resulting in over- or underestimation of postoperative PFT values, particularly for DLCO, which is sensitive to diffusion and microvascular abnormalities (15,30,31).

The clinical implications are straightforward, superior predictive accuracy can refine procedure selection (segmentectomy *vs.* lobectomy), risk assessment, patient counseling, and eligibility among borderline candidates (32,33). This is especially important since oftentimes patients are referred to surgeons with preoperative workups already completed beforehand. In the settings of borderline results where decision on the surgical eligibility depends on the minimal deviations in the numeric data, SPECT/CT brings a more efficient diagnostic strategy allowing streamlined and optimized referrals to surgical care. Systematic discrepancies observed with QVQ, especially for DLCO, may inappropriately exclude acceptable surgical candidates from curative resection or underestimate risk in patients with limited pulmonary reserve (34,35). In the circumstances where SPECT/CT availability is limited, ASC remains an acceptable option, but its proportional bias at higher values and vulnerability to parenchymal heterogeneity must be recognized.

Per the American College of Chest Physicians guidelines, cardiopulmonary exercise testing derived VO_2max is a stronger global predictor of operative risk than FEV_1 and DLCO and should be considered when simple functional tests are limited (8). This provides context that, particularly in patients with more preserved PFTs, systemic fitness determinants may drive more risk than isolated pulmonary metrics (36). Accordingly, reduced discrimination of ASC at higher values may reflect simplified anatomical assumptions, whereas SPECT/CT retains an advantage by quantifying

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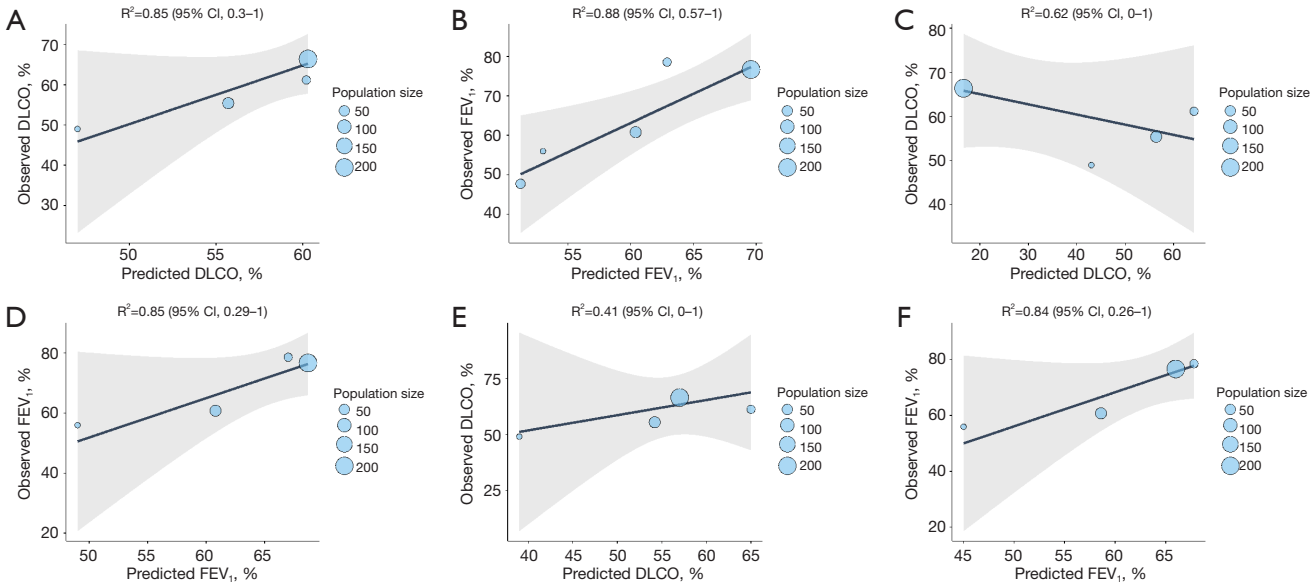


Figure 3 Linear regression models demonstrated the concordance between predicted and observed values for SPECT/CT (A,B), QVQ (C,D), and ASC (E,F). SPECT/CT showed the strongest correlation for both DLCO and FEV_1, with R^2 values approaching 1, indicating strong predictive accuracy. QVQ showed a moderate association, whereas ASC showed weaker performance for DLCO but similar performance for FEV_1. ASC, anatomical segment calculation; CI, confidence interval; DLCO, diffusing capacity for carbon monoxide; FEV_1, forced expiratory volume in one second; QVQ, quantitative ventilation/perfusion; SPECT/CT, single-photon emission computed tomography/computed tomography.

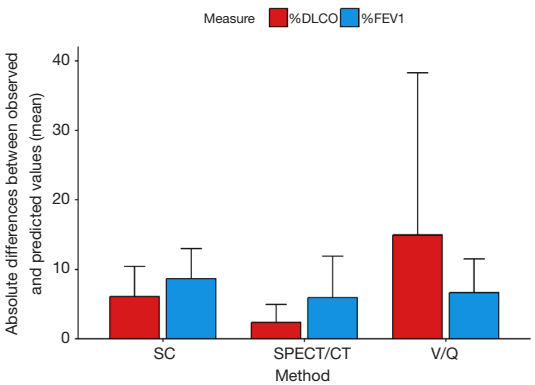


Figure 4 Absolute differences between predicted and observed values of FEV_1 and DLCO across diagnostic methods, taller bars indicate greater discrepancies and therefore inferior diagnostic performance. SPECT/CT demonstrated the lowest discrepancies, indicating the highest accuracy, whereas ASC showed intermediate accuracy and QVQ the lowest accuracy. ASC, anatomical segment calculation; DLCO, diffusing capacity for carbon monoxide; FEV_1, forced expiratory volume in one second; QVQ, quantitative ventilation/perfusion; SC, segment counting; SPECT/CT, single-photon emission computed tomography/computed tomography; V/Q, ventilation/perfusion.

regional function (10).

This meta-analysis adds to the field of thoracic surgery by head-to-head comparison across three modalities of predicting postoperative pulmonary function in patients, undergoing anatomic lung resection for cancer, leveraging complementary metrics (agreement, correlation, absolute error) and evaluating both FEV_1 and DLCO, two most important independent determinants of surgical decision-making. Beyond quantifying the magnitude of benefit, our results are internally consistent and suggest that the

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Figure 5 Critical appraisal of studies according to Quality Assessment of Diagnostic Accuracy Studies.

advantage is most pronounced for DLCO, where QVQ was especially imprecise.

This study has several limitations. First, the number of studies and overall sample size are modest, limiting subgroup analyses (e.g., extent of resection, COPD severity, ventilation *vs.* perfusion protocols). Second, methodological heterogeneity across centers (tracers, acquisition/reconstruction, segmentation/thresholding, timing of postoperative PFTs) may introduce uncontrolled variability. Third, all the reported data were observational, with inherent risks of selection bias and residual confounding. Lastly, differences in predicted postoperative values computation and in reporting of variance could influence estimates. To mitigate these issues, we emphasized head-to-head comparisons within-study, employed standardized agreement/correlation metrics, and evaluated two independent functional endpoints (FEV_1 and DLCO).

Conclusions

Among adults with NSCLC undergoing anatomic lung resection, preoperative SPECT/CT provides superior prediction of postoperative FEV_1 and DLCO values compared to that of QVQ scan and ASC, yielding smaller bias, tighter limits of agreement, and stronger correlations. SPECT/CT should be considered the preferred modality of preoperative patient evaluation where available. Future work should prioritize protocol standardization and prospective multicenter validation of decision thresholds.

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Footnote

Reporting Checklist: The authors have completed the PRISMA reporting checklist. Available at https://jtd.amegroups.com/article/view/10.21037/jtd-2026-1-0155/rc

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Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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